

Explore Quantum walk in Neuronal Network Modeling

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1 Block Matrix Formulation of Time Evolution

Let the original time evolution be given by $\eta_t = AY_{t-1} + B$, where the elements of B are bounded in $[0, b_{\max}]$ and the elements of Y are binary (0 or 1). We construct the extended block matrices A' and B' as:

$$A' = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix}, \quad B' = \begin{bmatrix} B \\ 0 \end{bmatrix} \quad (1)$$

To properly reconstruct η_t in the upper block, we must define Y'_{t-1} with the variables in the lower block:

$$Y'_{t-1} = \begin{bmatrix} 0 \\ Y_{t-1} \end{bmatrix} \quad (2)$$

The extended time evolution of η'_t then evaluates correctly:

$$\begin{aligned} \eta'_t &= A'Y'_{t-1} + B' \\ &= \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} 0 \\ Y_{t-1} \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} AY_{t-1} \\ 0 \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} AY_{t-1} + B \\ 0 \end{bmatrix} \end{aligned} \quad (3)$$

Substituting the original equation into the upper block, we arrive at the final compact form:

$$\eta'_t = \begin{bmatrix} \eta_t \\ 0 \end{bmatrix} \quad (4)$$

Finally we measure the upper half of qubits to retrieve η_t . Assume $b_{max} = 0.3$ and draw B,Y randomly.